

SHAWN NUGUID

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SUMMARY OF QUALIFICATIONS

- Third-year Computer Engineering student with hands-on experience in hardware design and validation, including PCB layout checks using KiCad, Verilog/SystemVerilog testbench development, and C/C++ firmware debugging
- Skilled in executing hardware test and debug activities on printed circuit boards, board programming, prototype inventory management, and embedded system validation
- Strong collaborator with a track record of improving test consistency and system reliability across complex embedded systems

TECHNICAL SKILLS

Programming Languages: C/C++ | Python | Java | VHDL | Verilog | SystemVerilog | SQL | JavaScript | TypeScript | R

EDA Tools & Software: Quartus II | KiCad | MATLAB | ModelSim | Visual Studio | Linux | Git | Bash/Shell | Cadence CAD Tools | Schematic Capture

Hardware & Prototype: Oscilloscope | Multimeter | Ammeter | Power Supplies | Routers | Switches | Arduino | RPi | Board Programming | PCB Test and Debug | Layout Review

Design & Validation: Hardware Design | Hardware Validation | Design Verification | Signal Integrity Analysis | Thermal Performance Analysis | Component Qualification | Design Documentation

EXPERIENCE

Pocket Clinic Inc.

Mar. 2025 – May 2025

Hardware Design and Validation Engineer

Toronto, ON

- Automated hardware design verification and validation of RFID-based I2C/SPI interfaces using **Verilog/SystemVerilog** testbenches, improving protocol coverage, test consistency, and issue tracking by **95%** across repeated validation cycles
- Developed and debugged **C-based firmware**, performed **board programming**, and built **Python** testbenches for smart injector operations and component degradation testing, using **MATLAB** and **Excel** to analyze results, identify problems, and improve debugging efficiency by **50%**
- Validated PCB schematics and layouts in **KiCad** using **ERC/DRC** checks while researching **ESP32** and **ATmega32** design considerations to support signal integrity analysis, modularity, and future hardware integration

I Challenge Diabetes Camp

Jul. 2025 – Sep. 2025

Software Developer

Toronto, ON

- Built a **Supabase SQL** UPSERT pipeline with **JSONB staging** to upload, validate, and structure camper records, improving database flow, data accuracy, and sync speed by **80%**
- Built and refactored HTTP REST APIs using **Node.js**, **Express**, **JavaScript**, and **TypeScript** for cloud-hosted applications, collaborating with cross-functional teams to ensure thorough design documentation and enabling scalable, real-time access to camper health data for over **200 parents** during camp
- Developed a **TypeScript + Supabase** CSV staging frontend using Papa.parse, implementing real-time validation rules to reduce upload errors, support cleaner data releases, and eliminate server downtime through continuous updates in Linux environments

TMU Intramural & Sport

May 2025 – May 2026

Intramural Tech and Program Lead

Toronto, ON

- Increased resume screening speed by **70%** by developing an **AI-powered ATS** using **Python**, **Pandas**, and **scikit-learn**, improving candidate prioritization through automated ranking and structured evaluation criteria
- Cleaned and maintained an internal database of over **16,000 users** using **Python** and **Excel**, removing outdated accounts to improve data accuracy, retrieval speed, and workflow efficiency while documenting processes for team collaboration
- Built **Tableau** dashboards for sports analytics, helping visualize participation trends, program performance, and operational decision-making

EDUCATION

Bachelor of Computer Engineering | Toronto Metropolitan University (TMU)

April 2028

Course Work: Microprocessor Systems, Control Systems, Computer Organization and Architecture, Digital Systems, Electric Circuit Analysis, Algorithms and Data Structures